

Requirement Analysis

Solution Requirements (Functional & Non-functional)

Date	8 October 2026
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Project Name	Metro Ticket Generating System in ServiceNow
Maximum Marks	

Functional Requirements

FR No.	Functional Requirement (Epic)	Sub Requirement
FR-1	Ticket Booking Catalog Item	Service Catalog item with variables Source, Destination, Passenger Type and Number of Tickets.
FR-2	Field Validation	Non-empty source/destination, fare match and approved travel date.
FR-3	Mandatory Fields	Enforced on client and server side to prevent incomplete submissions.
FR-4	Auto-Populate Logic	Default values (e.g., single or return journey) using catalog client scripts and UI policies.
FR-5	Fare Calculation	Fare is calculated automatically from stations, passenger type and number of tickets.
FR-6	QR Code Generation	A digital ticket with a QR code is generated instantly after booking.
FR-7	Station Details Mapping	Station details are mapped to the booking for fare and validation.
FR-8	Field Mapping	Process Automation Engine auto-maps and populates values entered by the user.
FR-9	Notifications	Booking confirmation sent through Flow Designer and Notifications.
FR-10	UI Pages & Testing	User-facing UI pages, followed by UAT and deployment.

Non-functional Requirements

NFR	Requirement	Description
NFR-1	Usability	Simple and fast booking form for all passengers.
NFR-2	Security	ServiceNow platform and role-based access controls.
NFR-3	Reliability	Validation rules consistently prevent incomplete or wrong bookings.
NFR-4	Performance	Normal interactive ServiceNow response for booking and QR generation.
NFR-5	Availability	Depends on the ServiceNow instance.
NFR-6	Scalability	Extendable with payments, analytics and WhatsApp booking.

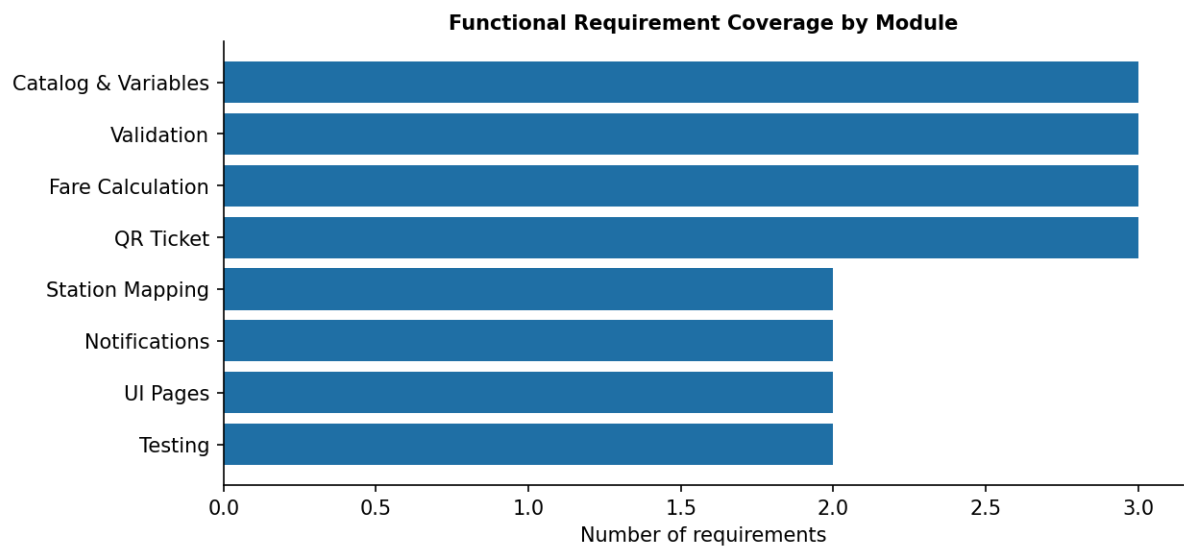


Figure 1: Functional requirement coverage.